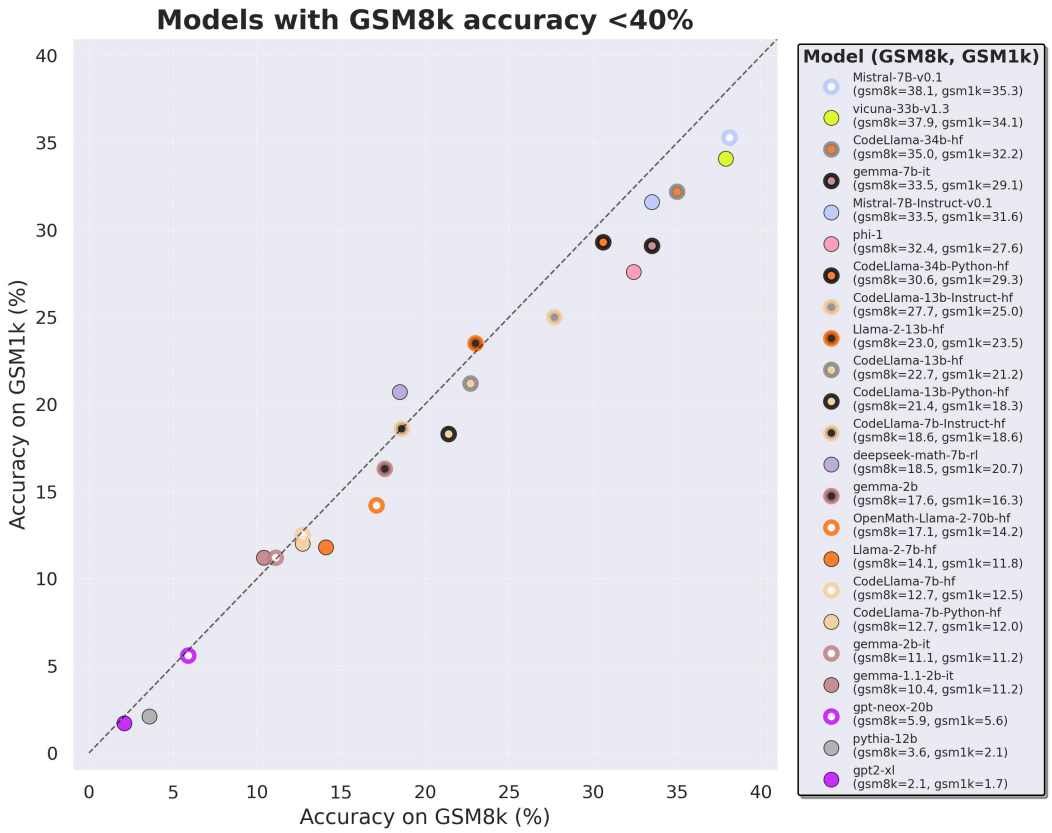

Figure 12: Models with between 0 and 40% accuracy on GSM8k compared to the line of no overfit. This plot is zoomed into the relevant sections (0-40% accuracy).

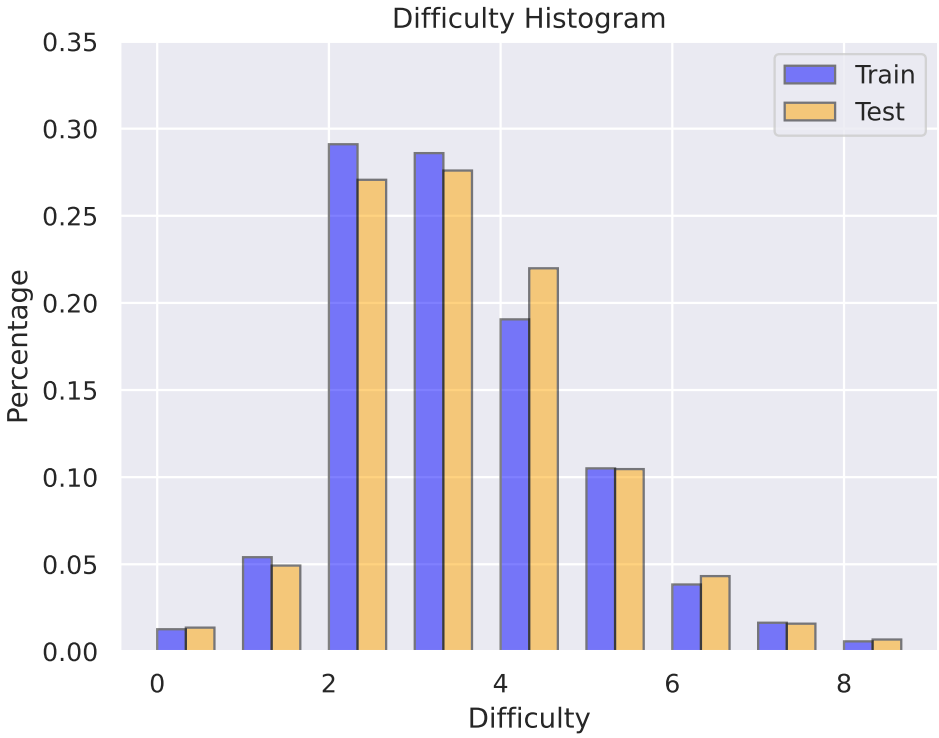

Figure 13: Approximate difficulty distribution of GSM8k train and test sets, measured by number of required steps to solve the problem. GSM1k annotators were instructed to create problems matching the overall distribution of the combined train and test difficulty distribution. The process of estimating problem difficulty is described in Section 3.2.

J Ablation with Human Answer Extraction

Because LM Evaluation Harness uses an automatic extraction format which takes the last number outputted as the model's final answer and compares an extract string match with the gold standard. Models answers which fail to follow the proper format may be marked as incorrect even if the model produced the "correct" answer. This most frequently occurs when the model phrases its answer in natural language, but adds extraneous information at the end of its sentence. Additionally, automatic extraction is brittle: in a few cases we observe that model outputs such as "24.0" are marked as not matching the gold standard answer of 24 due to the trailing decimal points failing the exact string match.

To measure the impact of extraction errors, we analyze a subset of models and use human annotators to extract model answers. These models were chosen for high performance on GSM1k for the purposes of creating a leaderboard of top performing models. While the absolute performance numbers change, we do not find meaningful differences in the amount of overfitting between GSM1k and GSM8k based on whether human or automatic extraction was used for this set of models.

Model	**GSM1k (Human)**	**GSM1k (Auto)**	**GSM8k (Human)**	**GSM8k (Auto)**	**Gap (Human)**	**Gap (Auto)**
Claude 3 Opus	0.952	0.825	0.955	0.802	-0.003	0.023
GPT-4 Turbo Preview	0.951	0.898	0.952	0.898	-0.001	0.0
GPT-4o	0.949	0.928	0.962	0.933	-0.014	-0.005
Claude 3 Sonnet	0.933	0.744	0.926	0.719	0.006	0.024
Gemini 1.5 Pro (post-I/O)	0.923	0.895	0.933	0.915	-0.01	-0.02
Gemini 1.5 Pro (pre-I/O)	0.905	0.885	0.904	0.897	0.002	-0.011
Llama 3 70B Instruct	0.901	0.895	0.917	0.896	-0.016	-0.001
Gemini 1.5 Flash	0.901	0.832	0.896	0.804	0.005	0.027
Mistral Large	0.875	0.853	0.892	0.853	-0.017	0.0
Gemini 1.0 Pro	0.798	0.789	0.805	0.792	-0.007	-0.002
CodeLlama 34B Instruct	0.375	0.366	0.422	0.415	-0.047	-0.049